

Day 24 Student Handout

Geometry Summer Credit | Coordinate Geometry Synthesis

Name:	Date:	Period:
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Learning Targets

I can use slope, distance, midpoint, and circle equations together. I can connect coordinate geometry to Algebra 2 ideas like transformations and conics. I can justify a geometric claim using coordinate evidence.

Lesson 24A: Coordinate Tools Working Together

Warm-Up

1.	Find the slope of the line through A(2, 5) and B(8, 17).
2.	Find the distance between C(-3, 4) and D(5, 10).
3.	Find the midpoint of E(-6, 9) and F(4, -1).

Coordinate Geometry Toolkit

Slope tests parallel and perpendicular lines. Distance tests length and congruence. Midpoint locates centers and bisectors. Circle equations connect center and radius.

Problem	Work / notes
A(1, 2), B(7, 2), C(7, 6), D(1, 6). Use coordinate evidence to classify ABCD.	Slopes: Side lengths: Conclusion:
Find the equation of a circle with center (3, -4) and radius 6.	Formula: Substitute: Equation:
A circle has equation $(x - 2)^2 + (y + 5)^2 = 49$. Name its center and radius.	Center: Radius:

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Lesson 24B | Algebra 2 Bridge: Conics and Transformations

From Geometry to Algebra 2

Bridge Idea

A circle is one type of conic section. In Algebra 2, you will also study parabolas, ellipses, and hyperbolas. Transformations shift graphs by changing the equation.

Problem	Work / notes
Compare $y = x^2$ and $y = (x - 4)^2 + 1$. Describe the transformation.	Horizontal shift: Vertical shift:
Compare $x^2 + y^2 = 25$ and $(x - 3)^2 + (y + 2)^2 = 25$. Describe the transformation.	Original center: New center: Shift:
A circle's center moves from $(0, 0)$ to $(-5, 3)$ but the radius stays 4. Write the new equation.	Center-radius form: Equation:

4.	Explain how completing the square helps turn a circle equation into center-radius form.
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5.	What coordinate tools would you use to prove that a quadrilateral is a rectangle?
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